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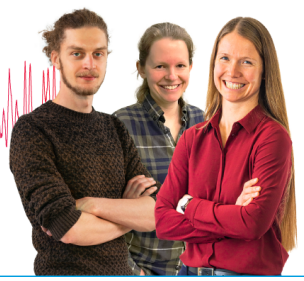
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Radiative transition probabilities for the $B\,^1\Pi$–$X\,^1\Sigma^+$ and $B\,^1\Pi$–$A\,^1\Sigma^+$ bands of ^{7}LiH[a)]

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The recently reported [J. Chem. Phys. **66**, 5412 (1977)] hybrid potential curves for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of LiH and the herein reported hybrid potential curve for the $B\,^1\Pi$ state of LiH are combined with the *ab initio* transition moments of Docken and Hinze [J. Chem. Phys. **57**, 4936 (1972)] to calculate radiative transition probabilities between all $B\,^1\Pi$ ($v' \leq 2$) and all $X\,^1\Sigma^+$ ($v'' \leq 23$) and $A\,^1\Sigma^+$ ($v'' \leq 26$) vibrational levels of ^{7}LiH. The strongest single emission band found is the $B\,^1\Pi$ ($v' = 2$)$\rightarrow X\,^1\Sigma^+$ ($v'' = 23$) band between the highest "long-range" levels in each potential. For all $B\,^1\Pi$ levels the emission is dominated by $B \rightarrow X$ rather than $B \rightarrow A$ emission.

I. INTRODUCTION

The simplest neutral heteronuclear molecule LiH has been the object of intense theoretical and spectroscopic interest[1] since the 1930's. We have embarked on a long-term project to fully understand the spectroscopy of the three lowest singlet states ($X\,^1\Sigma^+$, $A\,^1\Sigma^+$, and $B\,^1\Pi$) of LiH for a variety of reasons discussed elsewhere.[2]

As a step in this project, we have constructed full potential curves (Fig. 1) for the $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, and $B\,^1\Pi$ states of ^{7}LiH and from these curves calculated the vibrational (and vibrotational) energy levels of the isotopic lithium hydrides.[2] In this paper we summarize our calculations for the radiative transition probabilities between the three vibrational levels of the $B\,^1\Pi$ ($0 \leq v' \leq 2$) state and all vibrational levels of the $X\,^1\Sigma^+$ ($0 \leq v'' \leq 23$) and the $A\,^1\Sigma^+$ ($0 \leq v'' \leq 26$) states of ^{7}LiH. A more complete compilation of the results of these calculations is also available.[3] Other publications will report ^{7}LiH calculations of the lifetimes of all vibrational levels in the $A\,^1\Sigma^+$ state,[4] the transition probabilities of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ bands,[5] the lifetimes of all vibrational levels in the $B\,^1\Pi$ state,[6] the ir transition probabilities for the $X\,^1\Sigma^+$ state,[7] and a detailed examination of the "long-range molecules"[8] corresponding to $X\,^1\Sigma^+$ ($v = 23$), $A\,^1\Sigma^+$ ($v = 26$), and $B\,^1\Pi$ ($v = 2$).[9]

In the present manuscript we first examine the appropriate formulas for absorption and emission transition probabilities in the B–X and B–A bands of LiH in Sec. II. In Secs. III and IV we discuss construction of the $B\,^1\Pi$ potential and of the B–X and B–A dipole strength functions (electronic transition dipole moment as a function of internuclear distance), respectively. Results of the transition probability calculations are summarized in Sec. V. In Sec. VI we discuss the physical interpretation of these results.

II. RADIATIVE TRANSITION PROBABILITIES

There are four basic kinds of spectroscopic intensity measurements, depending on whether absorption or emission is involved, and whether photon or energy flux

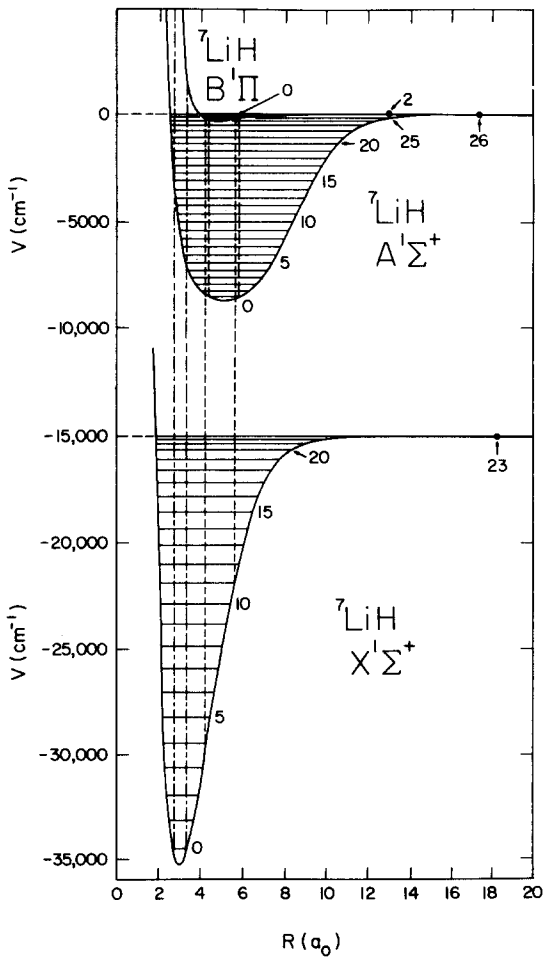


FIG. 1. Potential energy curves from Ref. 2 and this work for the $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, and $B\,^1\Pi$ states of ^{7}LiH, including calculated vibrational levels $v'' = 0$–23, $v'' = 0$–26, and $v' = 0$–2, respectively. The Franck–Condon regions for the various zero-point levels are indicated.

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TABLE I. The potential $B1$ (G in Ref. 11) consists of three regions, with the potential and its derivative being continuous at the boundaries.

Region	Range of $R(a_0)$	Function (a.u.)	Values
I	$R \leq 3.796253$	$A_0 + A_1 \exp(-A_2 R)$	a
II	$3.796253 < R < 17$	ICRKR (tensioned spline fit)	b
III	$17 \leq R$	$D_e - C_6 R^{-6} - C_8 R^{-8}$	c

[a] $A_0 = 2.858171 \times 10^{-6}$ a.u.; $A_1 = 48\,644.40131$ a.u.; $A_2 = 4.6295881 a_0^{-1}$.
[b] See Table II.
[c] From Ref. 11: $D_e = 1.3135107 \times 10^{-3}$ a.u. (288.28 cm^{-1}), $C_6 = 84.327$ a.u., $C_8 = 14794.02$ a.u.

is detected. The absolute intensities in each case involve a power of the transition frequency $\nu_{v',J';v'',J''} = h^{-1}[E'(v', J') - E''(v'', J'')]$ times the square of an electronic transition dipole matrix element

$$D_{v',J';v'',J''} \equiv \int_0^\infty \psi_{v',J'}(R) D(R) \psi_{v'',J''}(R) dR$$

[where the upper and lower state wavefunctions $\psi_{v',J'}$ and $\psi_{v'',J''}$ are normalized and where $D(R)$ is the dipole strength function (electronic transition dipole moment as a function of internuclear distance), discussed below in Sec. IV]. For the case of absorption measured by a decrease in photon flux the transition probability is proportional to $|D_{v',J';v'',J''}|^2$ (no frequency dependence) and commonly expressed as the line strength S or Einstein B coefficient. For absorption measured by a decrease in energy flux the transition probability is proportional to $|D|^2\nu$; the absorption oscillator strength f or absorption coefficient α is commonly used in that case. For emission the transition probabilities go as $|D|^2\nu^3$ and $|D|^2\nu^4$ for photon and energy detectors, respectively; the Einstein A coefficient is commonly used in the former case. An excellent discussion of the different quantities with conversion factors is given by Wiese, Smith, and Glennon.[10] A final quantity of interest is the Franck–Condon factor

$$q_{v',J';v'',J''} = \left| \int_0^\infty \psi_{v',J'}(R) \psi_{v'',J''}(R) dR \right|^2,$$

which is proportional to $|D|^2$ when $D(R)$ does not change significantly with R in the regions of appreciable integrand.

We shall restrict our attention here to $J' = J'' = 0$ only (although we have some limited $J > 0$ results), i.e., we shall ignore rotation, and report vibrational transition probabilities. Note that relatively strong rotational effects are expected, especially for $v' = 2$, but we do not discuss them here. We shall here report only Franck–Condon factors $q_{v',v''}$, squares of transition dipole matrix elements $|D_{v',v''}|^2$ [in units of $(e$ Å$)^2$], and spontaneous photon emission intensities $I(v', v'')$ [in units of $(e$ Å$)^2$ (cm$^{-1})^3$]. A more complete listing[3] and conversion factors[3,10] {e.g., $A_{v'v''}$ (sec^{-1}) $= \frac{1}{2} \times 7.2356 \times 10^{-6} I(v', v'')$ [$(e$ Å$)^2$ (cm$^{-1})^3$] for $\Pi \rightarrow \Sigma$} are given elsewhere. The transition frequencies $\nu_{v',v''}$ are available using the energies listed in Ref. 2 and in Sec. III.

TABLE II. Turning points $R_{v\pm}$ (in a_0) used for the potential $B1$ (Table I) from the isotopically combined Rydberg–Klein–Rees (ICRKR) potential for a hypothetical molecule of reduced mass $\mu = 100\ \mu_{^7\mathrm{LiH}}$ as a function of energy $E(v)$ in a.u. The underlined figures are not fully significant, but rounding would adversely affect interpolation.

v	$E(v)$ (10^{-6} a.u.)	$R_{v-}(a_0)$	$R_{v+}(a_0)$
$-\frac{1}{2}$	0	4.48974	
0	53.4504	4.27685	4.76103
1	157.2209	4.15244	4.82188
2	256.5946	4.08037	5.20204
3	351.3572	4.02906	5.38315
4	441.3130	3.98941	5.55848
5	526.3251	3.95723	5.73290
6	606.2887	3.93021	5.90953
7	681.1402	3.90690	6.09080
8	750.8658	3.88637	6.27879
9	815.4838	3.86806	6.47552
10	875.0487	3.85174	6.68313
11	929.6746	3.83767	6.90408
12	979.5119	3.82635	7.14084
13	1024.7699	3.81757	7.39491
14	1065.7177	3.81041	7.66663
15	1102.6650	3.80357	7.95522
16	1135.9627	3.79625	8.26050
17	1165.9024	···	8.58534
18	1192.7064	···	8.93577
19	1216.5416	···	9.32115
20	1237.4778	···	9.74560
21	1255.5518	···	10.22988
22	1270.8160	···	10.79249
23	1283.3737	···	11.46044
27.5	1296.8803	···	12.27393

TABLE III. Vibrational energy levels (in cm^{-1}) of the potential $B1$ (G in Ref. 11). The unphysical value $J = 0$ is used throughout.

v	$E(v)$
0	0.00
1	130.76
2	175.73
Dissoc.	179.04

III. POTENTIAL ENERGY CURVE FOR $B\,^1\Pi$ LiH

The potential energy curve for the $B\,^1\Pi$ state has previously been partially described (potential G of Ref. 11) and used in the course of determination of precise ^{7}LiH and ^{7}LiD dissociation energies.[11,12] In this work we relabel this potential $B1$, to be consistent with the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ potentials[2] used here also and labeled $X1$ and $A1$, respectively. For a plot showing this potential in greater detail see Fig. 1 of Ref. 11.

The $B1$ potential consists of three regions, which are given in Table I. In addition to the short-range exponential and long-range inverse power regions, a tensioned spline fit of the isotopically combined Rydberg–Klein–Rees (ICRKR) potential[11,13] given in Table II is used in the bulk of the potential well. For further discussion see Sec. II of Ref. 11. The eigenvalues for $J = 0$ in the $B1$ potential are given in Table III.

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TABLE IV. The dipole strength functions $D_{BX}(R)$ and $D_{BA}(R)$ in atomic units for the $B\,^1\Pi$–$X\,^1\Sigma^+$ and $B\,^1\Pi$–$A\,^1\Sigma^+$ transitions in ^{7}LiH.

	D_{BX}		D_{BA}		
Region	Range of $R(a_0)$	Function	Range of $R(a_0)$	Function	Reference
I	$R \le 1.0$	0.5950	$R \le 2.0$	−2.5837	Text
II	$1.0 \le R \le 12.0$	Tensioned spline	$2.0 \le R \le 12.0$	Tensioned spline	Table V, 14
III	$12.0 \le R$	3.3715	$12.0 \le R$	0.0434	14

IV. THE DIPOLE STRENGTH FUNCTION

The dipole strength functions $D_{BX}(R)$ and $D_{BA}(R)$ (i.e., the electronic transition dipole moments as a function of internuclear distance) have been constructed in a manner similar to that used previously for potential curves (e.g., in Sec. III and Ref. 2). The $D(R)$ functions each consist of three regions, given in Table IV and Fig. 2.

Region I assumes a constant value of 0.595 a.u. ($\sqrt{2} \times 0.421$) for $D_{BX}(R)$ for $R < 1.0$, for reasons given elsewhere,[5] and a constant value of −2.5837 for $D_{BA}(R)$ for $R < 2.0$, the innermost Docken–Hinze[14] point.

Region II is the region of predominant significance in our calculations. It consists of a cubic spline fit to the points given in Table V, which are the highly regarded *ab initio* results of Docken and Hinze.[14] As discussed elsewhere,[4] the excellent agreement (within mutual uncertainties of ~10%) between experimental and theoretical $A\,^1\Sigma^+$ lifetimes lends additional support for the related $D_{AX}(R)$ *ab initio* dipole strength function. It is our opinion that the $D(R)$ functions in the physically important region from 2–$12a_0$ are unlikely to be anywhere in error by as much as 20%. Finally, it should be noted that the change in these $D(R)$ functions with internuclear distance is far from negligible and thus simple approximations involving Franck–Condon factors or R centroids can not be expected to be quantitative.

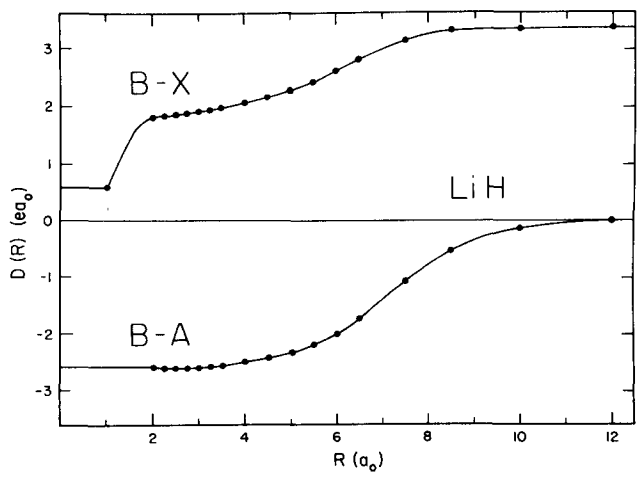


FIG. 2. Dipole strength function $D_{BX}(R)$ and $D_{BA}(R)$ (in atomic units) for the $B\,^1\Pi$–$X\,^1\Sigma^+$ and $B\,^1\Pi$–$A\,^1\Sigma^+$ transitions of ^{7}LiH. See Tables IV and V for details.

In asymptotic region III we have simply used the constant value of the last Docken–Hinze point.[14]

V. RADIATIVE TRANSITION PROBABILITY RESULTS

The Franck–Condon factors ($q_{v'v''}$, unitless), dipole strength function matrix element squares [$|D_{v'v''}|^2$, in $(e$ Å$)^2$ units], and spontaneous photon emission intensities [$I(v', v'')$, in $(e$ Å$)^2(\mathrm{cm}^{-1})^3$ units] as defined in Sec. II are given in Tables VI and VII for all vibrational levels v' of $B\,^1\Pi$ state and all v'' of the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of ^{7}LiH, respectively. The spontaneous emission intensities $I(v', v'')$ for both $B \rightarrow X$ and $B \rightarrow A$ band systems are also displayed in Fig. 3. A more complete listing is given elsewhere.[3]

These calculations have been carried out using an extensively modified version of the Franck–Condon factor program of R. N. Zare and co-workers,[15] based on the Numerov–Cooley numerical solution of the radial Schrödinger equation. Except for the higher levels (discussed elsewhere[9]), the calculations are typically

TABLE V. The dipole strength functions D_{BX} and D_{BA} (in a.u.) for the LiH B–X and B–A transitions in region II (see Table IV) consist of cubic spline fits to the points listed below. Except for the BX point at $1.0a_0$ discussed in the text, all points are from the work of Docken and Hinze.[14]

$R(a_0)$	$D_{BX}(R)$ (ea_0)	$D_{BA}(R)$ (ea_0)
1.0	0.5950	...
2.0	1.8222	−2.5837
2.25	1.8321	−2.6048
2.5	1.8570	−2.6096
2.75	1.8874	−2.6032
3.0	1.9223	−2.5870
3.25	1.9586	−2.5696
3.5	1.9960	−2.5445
4.0	2.0697	−2.4936
4.5	2.1612	−2.4269
5.0	2.2846	−2.3298
5.5	2.4342	−2.1949
6.0	2.6213	−1.9912
6.5	2.8221	−1.7220
7.5	3.1556	−1.0597
8.5	3.3051	−0.5209
10.0	3.3610	−0.1020
12.0	3.3715	0.0434

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TABLE VI. Franck–Condon factors ($q_{v'v''}$), dipole strength function matrix element squares [$|D_{v'v''}|^2$ in $(e\,\text{Å})^2$ units], and spontaneous photon emission intensities [$I(v',v'')$ in $(e\,\text{Å})^2\,(\text{cm}^{-1})^3$ units] for the $B\,^1\Pi$ ($v'=0$–2)–$X\,^1\Sigma^+$ ($v''=0$–23) bands of ^{7}LiH.

v' \ v''	0	1	2	3	4	5	6	7	8	9	10	11
						$q_{v'v''}$						
0	0.0010	0.0076	0.0247	0.0508	0.0792	0.1023	0.1171	0.1209	0.1140	0.0997	0.0820	0.0640
1	0.0008	0.0055	0.0160	0.0284	0.0358	0.0346	0.0262	0.0142	0.0040	0.0000	0.0042	0.0160
2	0.0002	0.0012	0.0035	0.0059	0.0068	0.0058	0.0036	0.0013	0.0000	0.0006	0.0027	0.0057
$\sum_{v'} q_{v'v''}$	0.002	0.014	0.044	0.085	0.122	0.143	0.147	0.136	0.118	0.100	0.089	0.086
						$\lvert D_{v'v''}\rvert^2$						
0	1.125(−3)	8.809(−3)	2.934(−2)	6.223(−2)	1.002(−1)	1.340(−1)	1.594(−1)	1.716(−1)	1.693(−1)	1.551(−1)	1.342(−1)	1.105(−2)
1	8.672(−4)	6.324(−3)	1.898(−2)	3.457(−2)	4.506(−2)	4.510(−2)	3.544(−2)	2.000(−2)	5.816(−3)	9.185(−7)	6.997(−3)	2.778(−3)
2	2.010(−4)	1.430(−3)	4.136(−3)	7.127(−3)	8.560(−3)	7.575(−3)	4.843(−3)	1.752(−3)	4.967(−5)	9.497(−4)	4.425(−3)	9.743(−3)
						$I(v',v'')$						
0	4.543(10	3.149(11)	9.291(11)	1.742(12)	2.476(12)	2.922(12)	3.062(12)	2.900(12)	2.517(12)	2.028(12)	1.542(12)	1.117(12)
1	3.543(10)	2.288(11)	6.084(11)	9.803(11)	1.129(12)	9.973(11)	6.908(11)	3.434(11)	8.787(10)	1.221(7)	8.182(10)	2.858(11)
2	8.245(9)	5.196(10)	1.332(11)	2.030(11)	2.155(11)	1.683(11)	9.485(10)	3.024(10)	7.545(8)	1.270(10)	5.205(10)	1.009(11)

v' \ v''	12	13	14	15	16	17	18	19	20	21	22	23	$\sum_{v'} q_{v'v''}$
						$q_{v'v''}$							
0	0.0476	0.0377	0.0228	0.0146	0.0089	0.0050	0.0025	0.0011	0.0004	0.0001	0.0000	0.0000	1.000
1	0.0332	0.0528	0.0718	0.0881	0.0999	0.1060	0.1053	0.0975	0.0818	0.0552	0.0218	0.0002	0.999
2	0.0085	0.0103	0.0106	0.0090	0.0060	0.0023	0.0000	0.0032	0.0217	0.0807	0.2599	0.4188	0.868
$\sum_{v'} q_{v'v''}$	0.089	0.101	0.105	0.112	0.115	0.113	0.108	0.102	0.104	0.136	0.282	0.419	
						$\lvert D_{v'v''}\rvert^2$							
0	8.710(−2)	6.572(−2)	4.747(−2)	3.264(−2)	2.118(−2)	1.275(−2)	6.891(−3)	3.191(−3)	1.135(−3)	2.247(−4)	8.548(−6)	4.067(−8)	
1	6.076(−2)	1.025(−1)	1.487(−1)	1.950(−1)	2.365(−1)	2.682(−1)	2.840(−1)	2.790(−2)	2.460(−1)	1.718(−1)	6.936(−2)	5.473(−4)	
2	1.548(−2)	1.999(−2)	2.178(−2)	1.987(−2)	1.408(−2)	5.863(−3)	7.507(−5)	9.075(−3)	6.477(−2)	2.500(−1)	8.222(−1)	1.333(0)	
						$I(v',v'')$							
0	7.744(11)	5.147(11)	3.280(11)	1.995(11)	1.150(11)	6.175(10)	3.000(10)	1.261(10)	4.121(9)	7.633(8)	2.778(7)	1.301(5)	
1	5.505(11)	8.190(11)	1.049(12)	1.218(12)	1.313(12)	1.329(12)	1.266(12)	1.130(12)	9.162(11)	5.990(11)	2.314(11)	1.798(9)	
2	1.411(11)	1.607(11)	1.547(11)	1.250(11)	7.870(10)	2.929(10)	3.375(8)	3.706(10)	2.434(11)	8.796(11)	2.768(12)	4.419(12)	

TABLE VII. Franck–Condon factors ($q_{v'v''}$), dipole strength function matrix element squares [$|D_{v'v''}|^2$ in $(e\,\text{Å})^2$ units], and spontaneous photon emission intensities [$I(v',v'')$ in $(e\,\text{Å})^2$ $(\text{cm}^{-1})^3$ units] for $B\,^1\Pi(v'=0\text{–}2)$–$A\,^1\Sigma^+(v''=0\text{–}26)$ bands of ^{7}LiH.

v' \ v''	0	1	2	3	4	5	6	7	8	9
						$q_{v'v''}$				
0	0.9618	0.0000	0.0278	0.0045	0.0022	0.0021	0.0004	0.0006	0.0001	0.0002
1	0.0129	0.3723	0.0838	0.1097	0.0860	0.0645	0.0586	0.0428	0.0376	0.0287
2	0.0001	0.0718	0.0050	0.0052	0.0011	0.0000	0.0004	0.0024	0.0044	0.0077
$\sum_{v'} q_{v'v''}$	0.975	0.444	0.117	0.119	0.089	0.067	0.059	0.046	0.042	0.037
						$\lvert D_{v'v''}\rvert^2$				
0	1.433(0)	7.579(−3)	2.912(−2)	1.938(−3)	7.522(−4)	1.109(−3)	2.645(−5)	2.309(−4)	2.990(−6)	4.331(−5)
1	3.352(−2)	4.858(−1)	5.197(−2)	7.387(−2)	4.328(−2)	2.316(−2)	1.864(−2)	8.734(−3)	6.823(−3)	3.430(−3)
2	8.623(−4)	9.655(−2)	2.175(−3)	3.736(−3)	5.408(−4)	1.604(−5)	9.118(−5)	7.174(−4)	7.329(−4)	1.109(−3)
						$I(v',v'')$				
0	8.356(11)	3.986(9)	1.362(10)	7.942(8)	2.664(8)	3.351(8)	6.724(6)	4.870(7)	5.155(5)	6.003(6)
1	2.048(10)	2.681(11)	2.556(10)	3.190(10)	1.619(10)	7.416(9)	5.038(9)	1.966(9)	1.261(9)	5.123(8)
2	5.352(8)	5.417(10)	1.088(9)	1.642(9)	2.062(8)	5.239(6)	2.516(7)	1.657(8)	1.387(8)	1.698(8)

v' \ v''	10	11	12	13	14	15	16	17	18	19
						$q_{v'v''}$				
0	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.0239	0.0187	0.0148	0.0116	0.0089	0.0070	0.0052	0.0040	0.0029	0.0022
2	0.0107	0.0145	0.0181	0.0216	0.0253	0.0288	0.0322	0.0358	0.0396	0.0438
$\sum_{v'} q_{v'v''}$	0.035	0.033	0.033	0.033	0.034	0.036	0.037	0.040	0.043	0.046
						$\lvert D_{v'v''}\rvert^2$				
0	1.103(−6)	4.147(−6)	1.095(−6)	5.079(−8)	7.240(−7)	1.156(−7)	3.626(−7)	1.146(−7)	9.766(−8)	3.156(−8)
1	2.348(−3)	1.224(−3)	7.032(−4)	3.898(−4)	1.770(−4)	1.021(−4)	3.610(−5)	2.049(−5)	4.494(−6)	1.934(−6)
2	1.040(−3)	1.006(−3)	9.250(−4)	7.505(−4)	6.000(−4)	4.175(−4)	2.934(−4)	1.703(−4)	9.581(−5)	3.772(−5)
						$I(v',v'')$				
0	1.207(5)	3.514(5)	7.022(4)	2.403(3)	2.450(4)	2.686(3)	5.587(3)	1.101(3)	5.459(2)	9.326(1)
1	2.787(8)	1.132(8)	4.968(7)	2.052(7)	6.746(6)	2.727(6)	6.484(5)	2.355(5)	3.109(4)	7.426(3)
2	1.268(8)	1.017(8)	6.751(7)	4.095(7)	2.380(7)	1.166(7)	5.548(6)	2.076(6)	7.108(5)	1.577(5)

v' \ v''	20	21	22	23	24	25	26	$\sum_{v'} q_{v'v''}$
						$q_{v'v''}$		
0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000[a]	0.0000[a]	1.000
1	0.0015	0.0011	0.0007	0.0004	0.0002	0.0001	0.0000[a]	1.000
2	0.0484	0.0535	0.0599	0.0702	0.0913	0.1371	0.1702[a]	0.999
$\sum_{v'} q_{v'v''}$	0.050	0.055	0.061	0.071	0.092	0.137	0.170	
						$\lvert D_{v'v''}\rvert^2$		
0	1.020(−8)	1.054(−9)	3.256(−10)	1.599(−9)	2.626(−9)	1.962(−9)	3.973(−10)	
1	1.160(−7)	9.847(−10)	1.243(−7)	2.030(−7)	1.178(−7)	7.493(−8)	5.236(−12)	
2	9.522(−6)	1.857(−9)	6.865(−6)	2.481(−5)	5.144(−5)	7.889(−5)	8.701(−5)	
						$I(v',v'')$		
0	1.400(1)	5.510(−1)	4.620(−2)	2.964(−2)	2.426(−4)	3.020(−3)[a]	2.740(−3)[a]	
1	2.224(2)	8.082(−1)	3.450(1)	1.255(1)	6.419(−1)	2.690(−4)	1.108(−6)[a]	
2	2.031(4)	1.754(0)	2.327(3)	2.119(3)	5.548(2)	1.728(1)	2.714(−1)[a]	

[a]Denotes an absorption rather than an emission process.

carried out with a 0.005 Å grid from 0.5–10.5 Å, and have been extensively checked for convergence with respect to grid spacing and inner and outer limits. We are confident of the *numerical* reliability of our q values to $< 10^{-4}$, $|D|^2$ values to $\lesssim 10^{-4}\ ea_0$, and our I values to $\lesssim 10^8\ (e\,\text{Å})^2(\text{cm}^{-1})^3$. Further discussion of the highest levels which present particular numerical difficulties will be provided elsewhere.[9]

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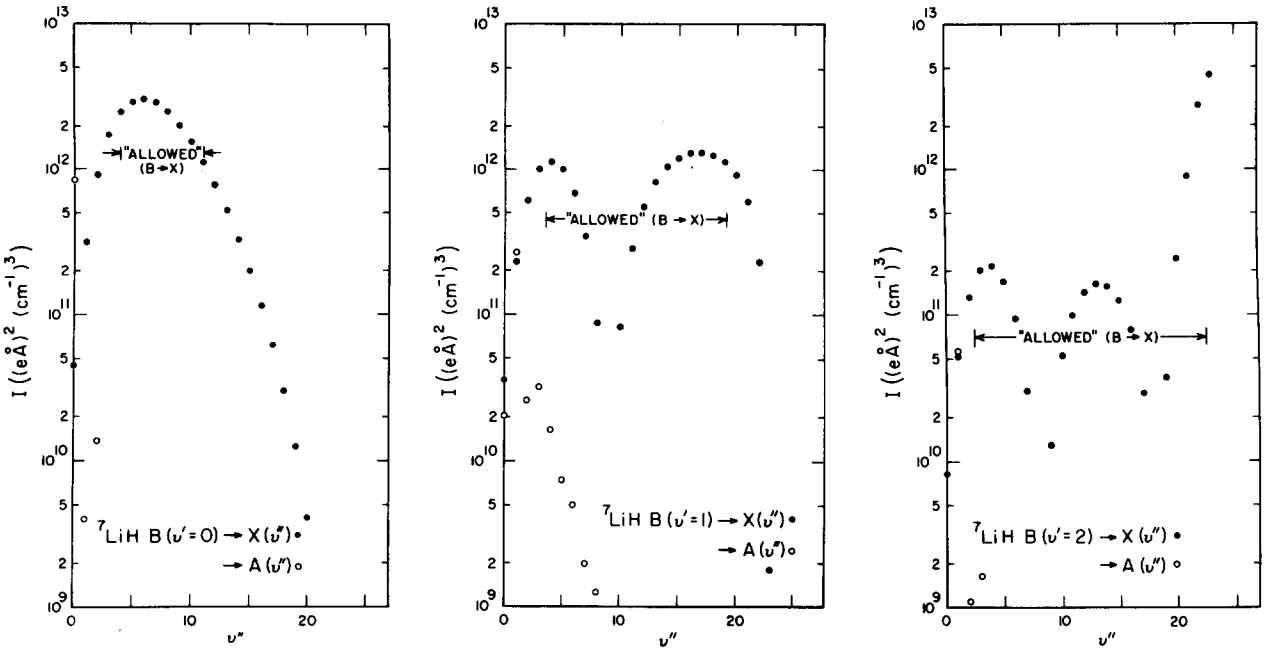


FIG. 3. Absolute spontaneous photon emission intensities $I(v', v'')$ [in $(e\,\text{Å})^2\,(\text{cm}^{-1})^3$] for the ^{7}LiH $B\,^1\Pi(v') \rightarrow X\,^1\Sigma(v'')$ (•) and $B\,^1\Pi(v') \rightarrow A\,^1\Pi^+(v'')$ (o) bands for (a) $v' = 0$, (b) $v' = 1$, (c) $v' = 2$, as a function of v''. Note the smooth behavior of $B \rightarrow X$ curves, with strong intensities only in the allowed regions (where $R_{v'-} \leq R_{v''+} \leq R_{v'+}$). Note also the B–X nodal dips (one in $v' = 1$ and two in $v' = 2$) where the internuclear distance of the node in the $\psi_{v'}$ wavefunction approximately equals $R_{v''+}$.

VI. DISCUSSION

The spontaneous photon emission intensities $I(v', v'')$ (Fig. 3) are worth examining in greater detail in terms of the potential energy curves involved (Fig. 1) ($q_{v',v''}$ or $|D_{v',v''}|^2$ could equally well be given). As shown in Fig. 1, emission for $v' = 0$ should occur to a variety of X and A state lower levels.

For the X state levels a rough criterion for a reasonably intense transition is that the outer turning points of the lower levels ($R_{v''+}$) be between the inner and outer turning points of the $v' = 0$ level ($R_{0'\pm}$) (see Fig. 1). The X state levels v'' satisfying this relationship ($R_{0'-} \leq R_{v''+} \leq R_{0'+}$) are termed "allowed" ($B$–$X$) in Fig. 3(a). The emission from $v' = 1$ is similar [Fig. 3(b)] except for a central "nodal" dip in intensity near $v'' = 8$. The somewhat expanded allowed region corresponds here to $R_{1'-} \leq R_{v''+} \leq R_{1'+}$. The central dip is associated with the node in the $v' = 1$ wavefunction. The emission from $v' = 2$ is similar [Fig. 3(c)] but with considerable expansion of the allowed region ($R_{2'-} \leq R_{v''+} \leq R_{2'+}$) and with two wavefunction nodes. Note that the intensities of allowed transitions decrease rapidly with increasing v' because of the increased number of vibrational levels to which emission occurs. For the highest v' of 2 (a long-range molecule), however, a new simplification occurs (discussed further elsewhere[9]) as most of the wavefunction amplitude is outside the outermost node and hence transitions to a similar level ($v'' = 23$) becomes the strongest molecular transition in the entire band system.

The experimental observations of absorption ($B \leftarrow X$)[16] do not yet include precise measurement of even relative intensities (no $B \rightarrow X$ emission observations exist). Nevertheless, it is clear that the strong absorption bands expected (all v' for $v'' = 0, 1$) are indeed observed.

For the A state levels the closeness of the inner and outer turning points of the B and A state zero-point levels implies very similar zero-point wavefunctions (overlap $= \sqrt{q} = 0.98$!) and a strong 0–0 transition (the strongest in the B–A band system), all other transitions being quite weak. For $v' = 1$ the 1–1 transition is still dominant and the strongest, but it is much weaker than the 0–0 transition as the $v' = 1$ and $v'' = 1$ turning points are now quite different. For $v' = 2$ there are no strong transitions, the largest overlap being 0.41 with the long-range level $v'' = 26$, but that level is actually slightly higher (not lower) in energy.

There are no observations in either absorption or emission of the B–A band system.

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